k	NQ	GSM8K	MATH-500	GPQA	AIME'24
2	22.4	15.6	14.0	3.0	0.0
5	22.8	46.4	35.0	5.1	0.0
10	22.2	69.0	56.0	12.1	16.7
15	22.2	76.6	63.0	10.1	23.3
20	21.8	77.8	69.0	11.1	26.7
25	22.0	78.0	72.0	10.1	26.7
30	21.4	78.4	74.0	11.1	26.7

Table 5: Effect of k on **accuracy** (%).

Token Usage. Larger k delays stopping and increases tokens; the curve flattens beyond $k \approx 25-30$.

k	NQ	GSM8K	MATH-500	GPQA	AIME'24
2	77.2	65.6	86.1	105.8	84.4
5	222.3	204.1	418.6	464.2	414.9
10	336.3	319.8	841.6	1138.6	1522.2
15	381.8	364.7	1177.1	1463.2	2507.7
20	402.0	379.1	1585.3	1754.0	3558.5
25	412.7	385.2	1676.5	1949.6	4247.2
30	420.1	396.1	1750.3	2098.8	4445.9

Table 6: Effect of k on **#tokens**.

Takeaway. $k \in [20, 30]$ yields near-saturated accuracy but lower efficiency; practical settings can choose k around $10-20$ to balance accuracy and token savings.

A.6 Effect of Early Stopping on Faithfulness

Early stopping enables models to generate predictions without completing the full reasoning trace, which raises concerns about the faithfulness of the generated rationale. Faithfulness, i.e., whether the reasoning chain logically supports the final answer, has been a key focus in recent studies (Lanham et al., 2023; Chen et al., 2025). To assess this aspect, we conduct a manual evaluation of reasoning chains generated by R1-Qwen-32B on GPQA under the Learn-to-Stop setting. We restrict our analysis to instances where the original model produced a correct answer and categorize the early-stopped generations into the following four classes:

- **Perfect Stop**: The model halts reasoning immediately after predicting the correct answer for the first time, and the preceding steps form a faithful chain of reasoning.
- **Late Stop**: The model already reaches the correct answer but continues with unnecessary reasoning, suggesting it could have stopped earlier.

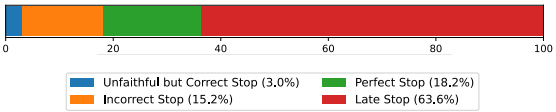

Figure 5: Faithfulness analysis of the Learn-to-Stop method on GPQA.

- **Incorrect Stop**: The model halts reasoning too soon, leading to an incorrect answer.
- **Unfaithful but Correct Stop**: The model outputs the correct answer, but the truncated reasoning trace does not support the answer logically. This is often caused by stopping before completing a faithful CoT.

We present the results in Figure 5. The majority of cases fall into the **Stop Late** category (63.6%), indicating that while the model reaches the correct answer, it continues with unnecessary reasoning, suggesting a suboptimal stopping point. This highlights the opportunity to further refine stopping criteria by incorporating finer-grained signals, e.g. hidden states from more layers that more precisely detect convergence and faithfulness in real time. The **Stop Perfectly** cases account for 18.2%, demonstrating the model's ability to identify the correct point to stop. Only 3% of cases are labeled as **Unfaithful**, where the correct answer is produced without supporting reasoning. This suggests that although our method may occasionally lead to unfaithful outputs, such instances are relatively rare in GPQA. Overall, these results indicate that our approach can reduce reasoning steps without significantly compromising the faithfulness of the generated reasoning. Future work could explore hybrid strategies that combine answer stability with faithfulness-aware signals, enabling models to adaptively balance efficiency and reasoning integrity across diverse tasks.

A.7 Test Cases from NQ

We list two examples from NQ in Table 7. In the first example, the original model initially generates the correct answer (Moira Kelly), but then adds unnecessary reasoning steps, including self-reflection, and revises the answer to an incorrect one (Julie Kavner). Similarly, in the second example, the model starts with the correct answer (Ashoka), but again adds unnecessary reasoning, ultimately revising it to an incorrect answer (Kalinga Nanda).

Question: who does the voice of nala in the lion king
Original: Okay, so I need to figure out who voices Nala in The Lion King. I remember watching the movie when I was a kid, but I can't recall the voice actress. Let me think. I know that in the original 1994 version, some of the main voices were well-known actors. Simba was voiced by Jonathan Taylor Thomas, Mufasa by James Earl Jones, and Scar by Jeremy Irons. But Nala... I'm not sure. Maybe it was someone like Jennifer Hudson? Wait, no, she was in the 2019 remake, I think. In the original, I think it was a different actress. Maybe someone like, I don't know, Moira Kelly? No, that doesn't sound right. Maybe it's someone else. Let me think of other voice actors from that era. Maybe it's someone like Eliza Dushku? Or maybe it's someone less famous. I think it's a British actress. Oh, maybe it's someone like, I don't know, Julie Kavner? Wait, no, she did Mufasa's sister in the original... **Correct Answer #**: Moira Kelly
Question: the king of kalinga in the mauryan period was
Original: Okay, so I need to figure out who the king of Kalinga was during the Mauryan period. Hmm, I remember that the Mauryan Empire was a significant empire in ancient India, and it was around the 3rd century BCE. The most famous king from that period is Ashoka the Great, known for his role in the Kalinga war.\n \n Wait, but the question is about the king of Kalinga, not the Mauryan king. So Ashoka was the Mauryan emperor, and he conquered Kalinga. I think the king of Kalinga at that time might have been a local ruler who resisted Ashoka's expansion. There's a specific name that comes to mind, something like Kalinga Nanda. I'm not entirely sure, but I recall that Nanda was a common name for kings in that region, and Kalinga Nanda was the ruler who opposed Ashoka.\n \n I should check if there are any other possible names. Some sources mention that the king of Kalinga was indeed Kalinga Nanda, and that Ashoka's conquest of Kalinga was a major event in his reign, leading to his conversion to Buddhism. So yeah, Kalinga Nanda is the name I'm looking for... **Correct Answer #**: Ashoka **Model:** R1-Qwen-32B

Table 7: Examples from NQ with responses from the original model.

A.7.1 Case Study

We provide a case study of the original model and the Learn-to-Stop method in Table 8.

A.8 Prompts

A.8.1 NQ

```
Answer the following question.
Directly output your final answer within \\boxed{}. DO
      NOT say anything else.


Question: {question}
```

A.8.2 Math Reasoning Tasks

```
Solve the following math problem.
Directly output your final answer within \\boxed{}. DO
      NOT say anything else.


Question: {question}
```

A.8.3 CCoT Prompt

```
Answer the following question.
You should think step by step, and limit the thinking
      process length to {LENGHT_BUGGED} words.
Directly output your final answer within \\boxed{}. DO
      NOT say anything else.


Question: {question}
```

```
Solve the following math problem.
You should think step by step, and limit the thinking
      process length to {LENGHT_BUGGED} words.
Directly output your final answer within \\boxed{}. DO
      NOT say anything else.


Question: {question}
```

Question: Each week, between 30 and 50 students show up for an archery class run by Betty and Wilma. Usually the students break up into groups of equal size for target practice. However, this week, Betty noticed that she could not break the students up into multiple groups of equal size. Wilma noticed that if she and Betty both joined the students in practicing, they still could not break the archers up into groups of equal size. How many students showed up to the archery class this week?
Original: Okay, so I have this math problem here about Betty and Wilma's archery class. Let me try to understand what it's asking.....47:\n 47 + 2 = 49. 49 is not prime because 7*7=49. So 47 is out.\n \nSo the only number that fits both conditions is 41. When you add 2, it becomes 43, which is also prime. Therefore, there must have been 41 students at the archery class this week. </think>\boxed{41} **Token #**: 741
Learn-to-Stop: Okay, so I need to figure out how many students showed up to the archery class this week. The problem says that each week, between 30 and ..... 41 fits both conditions. </think>\boxed{41} **Token #**:673
Dataset: MATH-500
Question: In a specific region of the sky, astronomers have observed that the number of stars varies with parallax as 1/plx^5. How does the number of stars in that region of the sky change with distance (per unit range of distance, r)?
Original: Okay, so I've got this problem here where astronomers observed that the number of stars varies with parallax as 1 over plx to the fifth power. Hmm, I need to figure out how the number of stars changes with distance, specifically per unit range of distance r..... Therefore, 5k / (1/r^6 * r) = 5k r^6 / r = 5k r^4. Yes, that's correct. So, the number of stars per unit distance is proportional to r^4. Wait, but that would mean that as you go further out, the number of stars per unit distance increases rapidly. That seems odd because usually, the number of stars in a volume element increases with the square of the radius (since volume is 4dr), but here it's increasing faster than that. Maybe because the number density itself is increasing with distance? Wait, no. The problem states that the number of stars varies with parallax as 1/plx^5. So, if parallax decreases (distance increases), the number increases as 1/plx^5, which is equivalent to d^5. So, the total number up to distance d is proportional to d^5. But the number per unit distance would require considering the derivative. Wait, maybe I'm confusing cumulative counts with differential counts. Let me clarify. If N(pl) is the total number of stars with parallax less than pl, then N(pl) = k / pl^5. But parallax less than pl corresponds to distances greater than d = 1/pl. So, N(d) = k d^5. But then the number of stars beyond distance d would be N_total - N(d) = k d^5. Hmm, that doesn't make much sense because as d increases, N(d) increases, meaning more stars are at larger distances, which is expected. But to find the number per unit distance, we take the derivative of N(d) with respect to d. So, dN/dd = 5k d^4. So, the number of stars per unit distance at distance d is proportional to d^4. Wait, but that would imply that the number density is increasing with distance, which might not align with standard stellar distributions unless there's an actual increase in star density with distance, which isn't typically the case. Normally, the number density of stars in the galaxy decreases with distance from the galactic center, but in this problem, we're probably considering a region where the distribution follows this specific law. So, putting it all together, the number of stars per unit distance (r) is proportional to r^4. </think>\boxed{r^4} **Token #**: 1952
Learn to Stop: Okay, so I have this problem where the number of stars varies with parallax as 1 over plx to the fifth power. Hmm, I remember that parallax is related to distance. Let me think about how they're connected.\n \n Wait, parallax (pl) is inversely proportional to distance (d) ..... Therefore, the number of stars per unit distance (N(r)) is proportional to r^4. </think>\boxed{r^4} **Token #**:1431
Dataset: GPQA-Diamond

Table 8: Examples from MATH-500 and GPQA-Diamond with responses from the original model and the Learn-to-Stop method. The original model generates redundant reasoning steps after producing the correct answer (41 and r^4), while the Learn-to-Stop method terminates reasoning earlier, resulting in a more concise response.